B. Tech CSE (Data Science)

SEMESTER –VIII

Sl No.	Course Code	Course Title	Hours Per Week			Total Credits
			Lecture	Tutorial	Practical	
1.	1008XX	Open Elective-II	3	0	0	3
2.	1008XX	Open Elective -III	3	0	0	3
3.	1538XX	Program Elective-V	3	0	0	3
4.	1538XX	Program Elective-VI	3	0	0	3
5.	100801P	Project-II	-	-	-	6
TOTAL						18

List of Open Elective Courses - II

sr.no.	Code	Course Title	L	T	P	Credit
1.	100833	Mobile Communications	3	0	0	3
2.	100834	Robotics	3	0	0	3
3.	100837	Digital Marketing and SEO	3	0	0	3
4.	100847	Data Communications and Systems	3	0	0	3
5.	100838	Evolutionary Computing	3	0	0	3

List of Open Elective Courses - III

sr.no.	Code	Course Title	L	T	P	Credit
1.	100848	Renewable Energy Systems	3	0	0	3
2.	100832	Cyber Laws and Rights	3	0	0	3
3.	100809	VLSI System Design	3	0	0	3
4.	100835	Optical Fiber Communication	3	0	0	3
5.	100819	Satellite Communication	3	0	0	3

List of Program Elective Courses - V

Sr. No.	Code	Course Title	L	T	P	Credit
1.	153801	Generative AI	3	0	0	3
2.	153802	AI in Cyber Security	3	0	0	3
3.	153803	Cyber Physical Systems	3	0	0	3
4.	153804	Quantum Computing	3	0	0	3
5.	153805	Blockchain Technology	3	0	0	3

List of Program Elective Courses - VI

Sr. No.	Code	Course Title	L	T	P	Credit
1.	153806	Game Theory	3	0	0	3
2.	153807	Cloud, Fog and Edge Computing	3	0	0	3
3.	153808	Industrial IoT	3	0	0	3
4.	153809	Object oriented DBMS (OODBMS)	3	0	0	3
5.	153810	Autonomous Drones	3	0	0	3

Semester	I	II	III	IV	V	VI	VII	VIII	Total
Credits	20.5	17.5	24	23	27	23	27	18	180

Semester-VIII

Open Elective-II

Course Code- 100833 Mobile Communications 3 0 0 3

Unit 1.0 6 hrs
An Overview of Wireless Systems: Introduction wireless communication.

Unit 2.0 8 hrs
Cellular System design concepts and fundamentals: Frequency Reuse – Channel Assignment -Handoff Strategies – Interference and System Capacity – Improving Coverage and Capacity in cellular systems.

Unit 3.0 8 hrs
Mobile Radio Wave propagation – I: Large scale path loss and propagation models – Reflection – Diffraction – Scattering – Practical link budget design – Outdoor propagation models – Indoor propagation models.

Unit 4.0 8 hrs
Mobile Radio Wave propagation – II: Small- Scale fading and multipath propagation, Rayleigh and Rician Distributions. Multiple Access Techniques for Wireless Communications -I – FDMA – TDMA –Spread Spectrum multiple access.

Unit 5.0 6 hrs
1st & 2nd Generation cellular Systems: AMPS; GSM features – Architecture – Radio subsystems – Traffic channels – call processing.

Unit 6.0 6 hrs
3rd & 4th Generation cellular Systems: CDMA features – Architecture – Forward and reverse channels –power control - system capacity; OFDM; 4G LTE systems.

Text/Reference:

1. William C Y Lee, Mobile Cellular Telecommunications, McGraw Hill (Main Book),2020, 2nd edition.
2. Stallings, Wireless Communications and Networks, Prentice Hall, 2004, 2nd edition.
3. Schwartz, Mobile Wireless Communications, Cambridge University Press (Main Book),2004.
4. Theodore S Rappaport, Wireless Communications Principles and Practice, Prentice Hall, 2001, 2nd edition.

Course Code- 100832 Cyber Laws and Rights 3 0 0 3

Unit 1.0 7 hrs

Cyber Security Fundamentals: Introduction, cybercrime and Information Security, Classification of Cybercrimes, Cybercrime the legal Perspective, Cyber offences,, Cyber-attacks, Social engineering, Cyber stalking, Botnets, Attack Vector, Cybercrime and cloud computing, Cybercrimes on mobile and wireless devices, Attacks on mobile/cell phones, Authentication service security, Organizational security policies and measures in mobile computing.

Unit 2.0 7 hrs

Ethics in Cyber Security: Privacy, Intellectual Property in cyberspace, Professional Ethics, Freedom of Speech, Fair User and Ethical Hacking, Trademarks, Internet Fraud, Electronic Evidence, forensic Technologies, Digital Evidence collection.

Unit 3.0 7 hrs

Tools and Methods Used in Cybercrime: Introduction, Proxy Servers and Anonymizers, Phishing, Password Cracking.

Unit 4.0 7 hrs

Types of threats: Key loggers and Spywares, Virus and Worms, Phishing and Identity Theft, Trojan Horses and Backdoors, Steganography, DoS and DDoS Attacks, SQL Injection, Buffer overflows.

Unit 5.0 8 hrs

Cybercrimes and Cyber security: Cybercrime and Legal Landscape around the world, Cyber laws, The Indian IT Act, Challenges, Digital Signatures and Indian IT Act, Amendments to the Indian IT Act, Cybercrime and punishment, Cost of Cybercrimes and IPR Issues, Web threats for Organizations, Social Computing and associated Challenges for Organizations.

Unit 6.0 6 hrs

Cybercrime Examples and Mini-Cases: Career Paths in Cyber security, Honeypots, Case study (Official Website Hacking, E-mail spoofing, Banking related Frauds, Credit Card related Frauds).

Text/Reference:

1. Cyber Security by Nina Godhole, Sunit Belapure, Wiley India.
2. Cyber Security Essentials by James Graham, Ryan Olson, Rick Howard CRC Press, Taylor & Francis Group, 2011 ISBN: 978-1-4398-5123-4.

Course Code- 153802 AI in Cyber security 3 0 0 3

Unit 1.0 8 hrs

Introduction to AI for Cyber security Professionals Applying Cloud Computing, Setting Up Your AI for Cyber security Arsenal Getting to know Python for AI and cyber security, Ham or Spam?

Unit 2.0 7 hrs

Detecting Email Cyber security Threats with AI, Malware Threat Detection Malware analysis at a glance.

Unit 3.0 5 hrs

Network Anomaly Detection with AI Network anomaly detection techniques.

Unit 4.0 8 hrs

Securing User Authentication Abuse Prevention, Fraud Prevention with Cloud AI Solutions Introducing fraud detection algorithms.

Unit 5.0 9 hrs

GANs - Attacks and Defenses GANs in a nutshell, Evaluating Algorithms Best practices of feature engineering, Assessing your AI Arsenal Evading ML detectors.

Unit 6.0 5 hrs

Artificial Intelligence tools and platforms in Cyber security.

Text/Reference:

1. Alessandro Parisi, Hands-On Artificial Intelligence for Cybersecurity: Implement smart AI systems for preventing cyber attacks and detecting threats and network anomalies, Packt Publication, August 2019, ISBN: 9781789804027.
2. L. F. Sikos, Ed., AI in Cybersecurity. Springer International Publishing, 2019. doi: 10.1007/978-3-319- 98842-9.
3. Shengjie Xu, Cybersecurity In Intelligent Networking Systems by, 2022, John Wiley.
4. James Graham, Richard Howard and Ryan Otson, Cyber Security Essentials, CRC Press, 2011.

Course Code- 153807 Cloud, Fog and Edge Computing 3 0 0 3

Unit 1.0 7 hrs

Demystifying the IOT Paradigm: The Enablement Platforms for IoT Applications and Analytics.

Unit 2.0 8 hrs

Cloud Computing Architecture: Overview of Cloud Architecture, Service Management in Cloud Computing. Cloud Service Models: IaaS, PaaS, SaaS Data Management in Cloud Computing, Resource Management in Cloud Computing, Cloud Implementation Strategies.

Unit 3.0 8 hrs

Open Source and Commercial Clouds: Overview and Comparison, Cloud Simulator and Research Trends in Cloud Computing.

Unit 4.0 7 hrs

Fog Computing: Introduction and Characteristics, VM Resource Allocation, Management, and Monitoring.

Introduction to Edge Computing: Fundamentals, Coordination of Cloud Services, The Cloud Computing Analytics Pipeline.

Unit 5.0 7 hrs

Serverless Computing and FaaS Model: Concepts and Applications, Cloud-Fog-Edge-enabled Analytics.

Cloud Security: Principles and Best Practices, Case Studies and Recent Advancements in Cloud Security.

Unit 6.0 5 hrs

Integration of Cloud and Edge: Collaborative Aspects, Interplay between Cloud, Fog, and Edge, Emerging Trends and Future Directions, Evolving Technologies (e.g., Quantum Computing), Social and Ethical Considerations in Cloud and Edge Computing.

Text/Reference::

1. Raj kumar Buyya,Satish Narayana Srirama, “Fog and Edge Computing: Principles and Paradigms”, 1st Edition, Wiley, 2019,978-1-119-52498-4
2. Pethuru Raj , Anupama C. Raman , "The Internet of Things: Enabling Technologies, Platforms, and Use Cases", 1st Edition, Taylor & Francis Books India Pvt. Ltd, 2022 ,978-1032509266
3. Rajkumar Buyya, James Broberg,Andrzej M. Goscinski, Cloud Computing: Principles and Paradigms, 1st Edition, Wiley, 2013 ,978-8126541256
4. Rajiv Misra, Yashwant Patel, Cloud and Distributed Computing: Algorithms and Systems, 1st Edition, Wiley, 2020,978-8126520275
5. Gautam Shroff, "Enterprise Cloud Computing - Technology, Architecture, Applications" , 1st Edition, Cambridge University Press, 2010 ,978-0521137355
6. Barrie Sosinsky, "Cloud Computing Bible", 1st Edition, Wiley, 2011 ,978-8126529803.
7. Ronald L. Krutz, Russell Dean Vines, "Cloud Security: A Comprehensive Guide to Secure Cloud Computing" , 1st Edition,Wiley, 2010 ,978-0470589878.
8. Nikos Antonopoulos, Lee Gillam, "Cloud Computing: Principles, Systems and Applications", 1st Edition, Springer London Ltd, 2012 ,978-1447125808.